Lecture 9: Comparing Two Population's Means

STAT 521: Applied Probability and Statistical Inference

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Independent Samples

- In many situations, we will select independent random samples from two populations to compare the population's parameters.
- Assume we sample n_1 observations from population 1 and n_2 observations from population 2.
- We will use the difference between the sample means $(\bar{y}_1 - \bar{y}_2)$ to make inference about $(\mu_1 - \mu_2)$

Sampling Distribution for $\bar{y}_1 - \bar{y}_2$

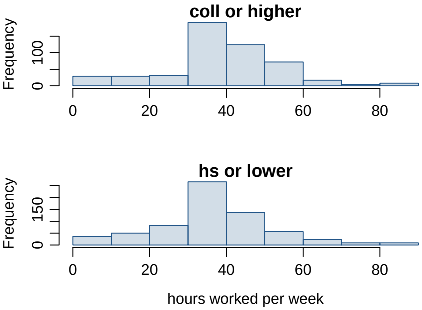
- The sampling distribution of $(\bar{y}_1 - \bar{y}_2)$ is approximately normal for large samples.
- The mean of the sampling distribution of $(\bar{y}_1 - \bar{y}_2)$ is $\mu_{\bar{y}_1 - \bar{y}_2} = \mu_1 - \mu_2$.
- The standard error of the sampling distribution of $(\bar{y}_1 - \bar{y}_2)$ is $\sigma_{\bar{y}_1 - \bar{y}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

$$\bar{y}_1 - \bar{y}_2 \sim N(\mu_1 - \mu_2, \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}})$$

Example

We want to construct a 95% confidence interval for the average difference between the number of hours worked per week by Americans with a college degree or higher and those with a high school degree or lower.

	$\bar{x}$	s	n
col or higher	41.8	15.14	505
hs or lower	39.4	15.12	667



Parameter and Point Estimate

- *Parameter of interest:* Average difference between the number of hours worked per week by *all* Americans with a college degree or higher and those with a high school degree or lower.

$$\mu_{col} - \mu_{hs}$$

- *Point estimate:* Average difference between the number of hours worked per week by *sampled* Americans with a college degree and those with a high school degree or lower.

$$\bar{x}_{col} - \bar{x}_{hs}$$

Checking assumptions & conditions

1. *Independence within groups:*
   - Both the college graduates and those with HS degree or lower are sampled randomly.
   - We can assume that the number of hours worked per week by one college graduate in the sample is independent of another, and the number of hours worked per week by someone with a HS degree or lower in the sample is independent of another as well.
2. *Independence between groups:* $\leftarrow$ *new!*
   Since the sample is random, the college graduates in the sample are independent of those with a HS degree or lower.
3. *Normality:*
   Both distributions look reasonably symmetric and/or the sample sizes are at least 30$\rightarrow$ we can assume that the sampling distribution of number of hours worked per week in both groups are nearly normal.

Confidence interval for difference between two means

- All confidence intervals have the same form:

$$point\ estimate \pm ME$$

- And all $ME = critical\ value \times SE\ of\ point\ estimate$
- In this case the point estimate is $\bar{x}_1 - \bar{x}_2$
- Since the sample sizes are large enough, the critical value is $z^\star$

- The only new concept is the Standard Error of the difference between two sample means:

$$SE_{(\bar{x}_1 - \bar{x}_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Let's put things in context

Calculate the standard error of the average difference between the number of hours worked per week by college graduates and those with a HS degree or lower.

	$\bar{x}$	s	n
col or higher	41.8	15.14	505
hs or lower	39.4	15.12	667

$$
\begin{aligned}
SE_{(\bar{x}_{col} - \bar{x}_{hs})} &= \sqrt{\frac{s_{col}^2}{n_{col}} + \frac{s_{hs}^2}{n_{hs}}} \\
&= \sqrt{\frac{15.14^2}{505} + \frac{15.12^2}{667}} \\
&= 0.89
\end{aligned}
$$

Confidence interval for the difference (cont.)

Using a 95% confidence interval, estimate the average difference between the number of hours worked per week by Americans with a college degree and those with a high school degree or lower.

$$\bar{x}_{col} = 41.8 \qquad \bar{x}_{hs} = 39.4 \qquad SE_{(\bar{x}_{col} - \bar{x}_{hs})} = 0.89$$

$$\begin{aligned}
(\bar{x}_{col} - \bar{x}_{hs}) \pm z^\star \times SE_{(\bar{x}_{col} - \bar{x}_{hs})} &= (41.8 - 39.4) \pm 1.96 \times 0.89 \\
&= 2.4 \pm 1.74 \\
&= (0.66, 4.14)
\end{aligned}$$

Note that both the lower limit and upper limit of the interval are positive. What does that say about the hours worked per week by Americans with a college degree compared to the ones with a high school degree or lower?

Interpretation of a confidence interval for the difference

Which of the following is the best interpretation of the confidence interval we just calculated?

(a) The difference between the average number of hours worked per week by college grads and those with a HS degree or lower is between 0.66 and 4.14 hours.

(b) college grads work on average 0.66 to 4.14 hours more per week than those with a HS degree or lower.

(c) *college grads work on average 0.66 to 4.14 hours more per week than those with a HS degree or lower.*

(d) college grads work on average 0.66 hours less to 4.14 hours more per week than those with a HS degree or lower.

(e) college grads work on average 0.66 to 4.14 hours less per week than those with a HS degree or lower.

Hypotheses Testing

What are the hypotheses for testing if there is a difference between the average number of hours worked per week by college graduates and those with a HS degree or lower?

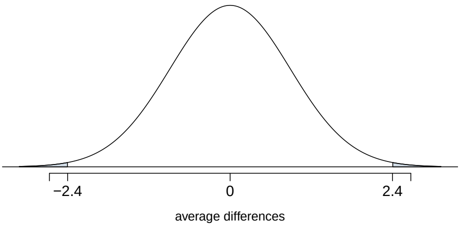
H_0: $\mu_{col} = \mu_{hs} \iff \mu_{col} - \mu_{hs} = 0$
There is no difference in the average number of hours worked per week by college graduates and those with a HS degree or lower. Any observed difference between the sample means is due to natural sampling variation (chance).

H_a: $\mu_{col} \neq \mu_{hs} \iff \mu_{col} - \mu_{hs} \neq 0$
There is a difference in the average number of hours worked per week by college graduates and those with a HS degree or lower.

Calculating the test-statistic and the p-value

$$(\bar{x}_{col} - \bar{x}_{hs}) \sim N\left((\mu_{col} - \mu_{hs}) = 0, SE_{(\bar{x}_{col} - \bar{x}_{hs})} = 0.89\right)$$

Note: The mean is 0 because in the null hypothesis we set μ_{col} equal to μ_{hs}, hence making the difference between them 0.



$$\begin{aligned} Z &= \frac{(\bar{x}_{col} - \bar{x}_{hs}) - 0}{SE_{(\bar{x}_{col} - \bar{x}_{hs})}} \\ &= \frac{2.4}{0.89} = 2.70 \\ upper\ tail &= 1 - 0.9965 = 0.0035 \\ p - value &= 2 \times 0.0035 = 0.007 \end{aligned}$$

Conclusion of the test

Which of the following is correct based on the results of the hypothesis test we just conducted?

(a) There is a 0.7% chance that there is no difference between the average number of hours worked per week by college graduates and those with a HS degree or lower.

(b) Since the p-value is low, we reject H_0. The data provide convincing evidence of a difference between the average number of hours worked per week by college graduates and those with a HS degree or lower.

(c) *Since the p-value is low, we reject H_0. The data provide convincing evidence of a difference between the average number of hours worked per week by college graduates and those with a HS degree or lower.*

(d) Since we rejected H_0, may have made a Type 2 error.

(e) Since the p-value is low, we fail to reject H_0. The data do not provide convincing evidence of a difference between the average number of hours worked per week by college graduates and those with a HS degree or lower.